

Muskaan Gauba

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EDUCATION

Master of Science, Computer Engineering

Arizona State University

Aug 2024 – May 2026

Tempe, Arizona

Coursework: Distributed Systems, Data Structures & Algorithms, OOP, Operating Systems, DBMS, Machine Learning

Bachelor of Technology, Computer Science and Engineering

Vellore Institute of Technology

Sept 2020 – May 2024

Vellore, India

Coursework: Data Structures, Algorithms, OOP, DBMS, Cloud Computing, Applied ML

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash

Backend: FastAPI, Node.js, REST APIs, GraphQL, gRPC, Microservices, System Design, Event-Driven Architecture, Observability

Frontend: React.js, Next.js, TypeScript, HTML, CSS, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Redis, Snowflake, FAISS, Pinecone

Cloud & DevOps: AWS (EC2, S3, SQS, SNS), Azure, Docker, Kubernetes, CI/CD, Prometheus, Grafana

Data & Streaming: Kafka, Spark, Airflow, dbt, ETL/ELT, Data Modeling

AI & ML: LangChain, LLMs, RAG, PyTorch, scikit-learn, XGBoost, MLflow, NLP

Tools: Git, GitHub, Jira, Agile, pytest, Jest, JWT, RBAC, OpenAPI, AI-Assisted Development

PROFESSIONAL EXPERIENCE

Arizona Department of Transportation (ADOT), Arizona State University

Oct 2025 – May 2026

Software Engineer – Graduate Research Assistant | Python, FastAPI, PostgreSQL, Docker, Kubernetes, AWS, LLMs

Tempe, AZ

- Built Python/FastAPI microservices processing 250K+ weekly records; integrated LLM APIs and AI-assisted code review tooling, automating manual review and cutting turnaround 75% across the service
- Redesigned PostgreSQL schema via composite indexing and table partitioning, cutting p99 query latency 39% across 5+ production services while supporting concurrent read-heavy workloads
- Automated CI/CD pipelines with Docker and Kubernetes on AWS, cutting release cycles 62% via zero-downtime rolling deployments and infrastructure-as-code provisioning
- Secured 10+ REST APIs with JWT auth and RBAC; sustained 85%+ pytest coverage, 100% SLA compliance, and full request-level observability in production

AssetPlus Consulting

Jan 2024 – Jul 2024

Software Engineer Intern | React.js, TypeScript, Node.js, REST APIs, Jest

Noida, India

- Built 15+ reusable React/TypeScript components with lazy loading and code-splitting, cutting page load time 25%
- Designed contract-first REST APIs with OpenAPI specs across 3+ modules, eliminating recurring integration errors entirely
- Shipped full-stack features across 4 Agile sprints with Jest unit/integration tests; zero critical production bugs

Mutelcor GmbH

Apr 2023 – Aug 2023

Software Engineer Intern | Python, Kafka, scikit-learn, Prometheus, Grafana

Delhi, India

- Scaled Kafka IoT pipeline to 500K+ daily events via consumer group partitioning, sustaining sub-50ms end-to-end latency
- Developed anomaly detection using Isolation Forest and Z-score scoring, achieving 95%+ precision and cutting MTTD 33%
- Implemented dead-letter queues, gRPC-based schema validation, structured logging across 8+ device types; eliminated message loss
- Reduced incident diagnosis time 40% via distributed tracing instrumented with Prometheus and Grafana

PROJECTS

Mem0-Style Memory Layer for AI Agents

Jan 2026 – May 2026

Python, FastAPI, PostgreSQL, Redis, FAISS, LangChain, Docker, AWS

- Engineered a long-term memory system for LLM agents combining vector search (FAISS), Redis key-value cache, and PostgreSQL relational store, enabling sub-100ms retrieval across sessions
- Designed REST APIs for memory read/write/delete with TTL expiry, namespace isolation, and embedding-based similarity search

Distributed Rate Limiter & API Gateway

Sept 2025 – Feb 2026

Python, Redis, FastAPI, Kubernetes, AWS EC2, Prometheus, Grafana

- Designed and implemented a system supporting token bucket and sliding window algorithms with Redis atomic ops, handling 50K+ RPS at sub-5ms p99
- Deployed as horizontally scalable gateway on Kubernetes with Redis cluster replication; 99.95% uptime across node failures

Real-Time Financial Data Pipeline & Analytics Dashboard

Aug 2025 – Dec 2025

Kafka, Spark, PostgreSQL, dbt, Airflow, AWS S3, Grafana

- Built Kafka + Spark Structured Streaming pipeline ingesting 100K+ market events/hour, orchestrated with Airflow DAGs and modeled with dbt star schema on PostgreSQL
- Designed Grafana dashboards tracking 15+ KPIs with freshness SLA and automated data quality validation at every pipeline stage

CERTIFICATIONS

Google AI Professional Certificate – Google / Coursera

May 2026